
IMPROVING DEEP LEARNING-BASED WILDFIRE SMOKE PLUME DETECTION WITH A MULTI-MODEL ENSEMBLE APPROACH

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ABSTRACT

With the increasing frequency and severity of wildfires, there is an urgent need for effective and rapid wildfire and smoke detection tools. Recent advancements in computer vision have demonstrated the potential of deep learning models, particularly neural networks, to automate the partitioning of high-resolution images into labelled segments. However, single-model approaches can struggle with generalization and accuracy in diverse conditions. To address these challenges, we propose creating

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an ensemble of deep learning models to produce more accurate annotations of wildfire smoke plumes and their relative density (light, medium, heavy) in Geostationary Operational Environmental Satellite (GOES) imagery. Our results indicate that ensemble techniques can improve performance compared to using a single model. This work builds multi-model ensembles that are expected to support fire and hazard management by being able to automate the monitoring of smoke in real-time from satellite imagery. Broadly, this will be a valuable tool for air quality and fire hazard management in the face of worsening wildfires.

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1 INTRODUCTION

Increased wildfire activity in recent years has led to increased smoke and particulate matter in the atmosphere, posing greater risks of respiratory illnesses and other air quality-induced health issues (Burke et al., 2021). Effective and timely wildfire and smoke detection tools are thus essential for supporting hazard management and mitigating risks to human health.

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The National Oceanic and Atmospheric Administration (NOAA) Geostationary Operational Environmental Satellites (GOES) provide high spatial and temporal resolution imagery of North America (Goodman et al., 2019), which can be leveraged to detect the presence and density of smoke plumes. The NOAA Hazard Mapping System (HMS) Fire and Smoke Product currently relies on human analysts to annotate the presence of smoke over North America using GOES imagery (McNamara et al., 2004). However, this product is limited by the availability of human analysts and their time. Specifically, annotations are outputted only once to several times a day and usually have a delay between smoke occurrence and the annotation. To address these limitations, we are leveraging advancements in deep learning to automate the detection of smoke from GOES imagery. Deep learning models, particularly encoder-decoder neural networks, have shown promise in automating the semantic segmentation (labelling images on a pixel-wise basis with multiple classes) of high-resolution images (Minaee et al., 2022). By automating this task, we can enable more frequent and consistent detection of smoke plumes.

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This proposal focuses on enhancing the capability of deep learning models to detect smoke through the use of multi-model ensemble techniques. It has been shown for classification tasks that ensemble methods, which combine the predictions of multiple classifiers, can of-

ten perform better than a single classifier (Dietterich, 2000). Particularly, utilizing a diverse set of classifiers in an ensemble is important to achieve the improvement in performance (Kuncheva and Whitaker, 2003). Furthermore, when using neural networks, combining the predictions of multiple independently-trained models can improve generalization and detection accuracy (Hansen and Salamon, 1990), (Cheng Ju and van der Laan, 2018), (Giacinto and Roli, 2001). This approach aims to provide a more reliable and accurate tool for real-time monitoring of smoke, ultimately informing fire and hazard management efforts and contributing to climate resilience and adaptation strategies.

2 DATA AND METHODS

The dataset we use consists of 183,672 samples, each with three spectral channels (C01-C03) of GOES imagery paired with HMS smoke annotations (pixel-wise labels of smoke density of light, medium, or heavy) for a specific time and location. The data spans 2018-2024, and we use 2023 for validation and 2022 for testing, with the remaining years used for training.

We utilize a variety of pre-developed encoder-decoder architectures that were designed for semantic segmentation contained within the Segmentation Models Pytorch library (Iakubovskii, 2019). These architectures include different features such as multi-scale fields-of-view and precise boundary detection (Chen et al., 2018), (Li et al., 2018), (Zhou et al., 2018), which are important for accurately detecting smoke plumes that can vary in size and appearance. Additionally, we select the best-performing single architecture and trained it with 12 different seeds to generate different initial random weights. These models are trained independently for 24 hours on 8 Nvidia P100 GPUs using the Adam optimizer, a learning rate of 0.001, a binary cross entropy loss function, and batch size of 128. After training, each model is selected based on its best validation IoU score.

The ensemble method we are using is an unweighted average of the model outputs. This method is straightforward to implement and has been shown to be effective in practice (Cheng Ju and van der Laan, 2018). This ensemble framework is shown in Figure 1. To explore how performance improves with a variety of model combinations, we test many ensemble sizes (1-12 models) for both combinations of model architectures and initial seeds.

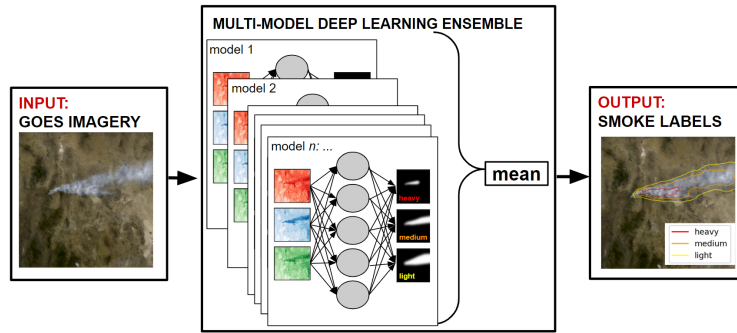


Figure 1: Multi-Model Ensemble Framework.

3 RESULTS

We measure model performance in terms of Intersection over Union (IoU) score (Equation 1) which quantifies the alignment between the model prediction (y_i^*) and the ground truth (y_i). Table 1 shows the IoU scores for individual models and ensembles. The ensemble of 8 different architectures outperforms the individual models, with an improvement in the IoU score over all densities and for each density individually. The ensemble of 8 models (with the same architecture, PAN) with different initial weights also outperforms the individual models, with a similar improvement in the IoU scores. Figure 2 shows the IoU performance

over all smoke densities as a function of ensemble size for the two ensemble schemes. The ensemble of with different initial weights generally improves as models are added to the ensemble. This improvement is likely due to the different initializations leading to the models searching different parts of the parameter space and thus finding different minima of the loss function. The ensemble of different architectures improves with more models up to 8 models, but then starts to decrease in performance. This decrease in performance could be due to the additional architectures not being as well suited for the task, or the additional models not having enough error diversity to improve ensemble performance. Figure 3 shows an example of smoke plume detection from the testing dataset. The ensemble predictions show smoother boundaries, making the prediction more comparable to the human-drawn polygon annotations.

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$$\text{IoU}_{\text{overall}} = \sum_{i=\text{light}}^{\text{heavy}} |y_i \cap y_i^*| \div \sum_{i=\text{light}}^{\text{heavy}} |y_i| \cup |y_i^*| \quad (1)$$

	Heavy	Medium	Light	Overall
Single Model: DLV3P	0.347	0.441	0.666	0.599
Single Model: PAN	0.349	0.478	0.664	0.604
Architecture Ensemble (N=8)	0.400	0.507	0.692	0.635
Random Initial Weights Ensemble (N=8)	0.409	0.512	0.684	0.631

Table 1: IoU results across three classes of smoke (light, medium, heavy) and over all densities. Presented for different individual models of different architectures ((Chen et al., 2018); (Li et al., 2018)), along with the architecture-based ensemble and random initial weights ensemble performance, where N denotes the number of models in the ensemble.

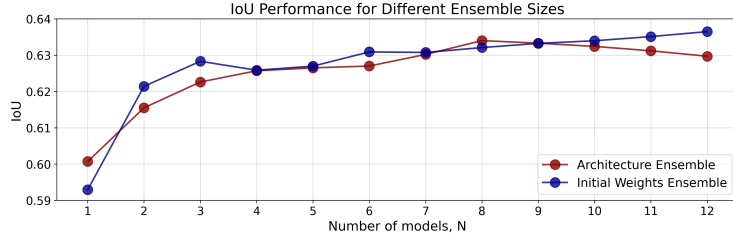


Figure 2: Ensemble IoU over all smoke densities as a function of ensemble size for two ensemble design schemes: random initial weights (blue) and architecture-based (red).

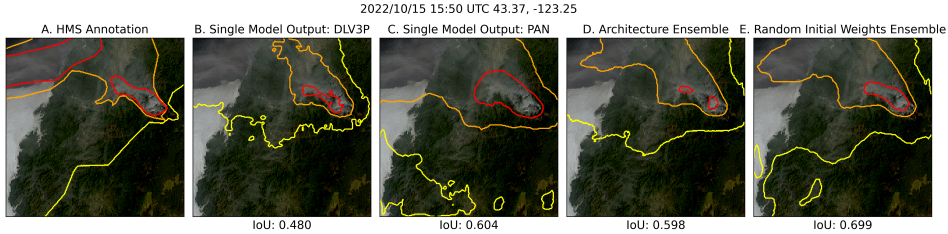


Figure 3: Example of smoke plume detection. Red contours outline the heavy density smoke, orange contours outline the medium density smoke, and yellow contours outline the light density smoke annotations.

4 CONCLUSIONS AND FUTURE WORK

We explore two schemes for building ensembles of deep learning models that both improve on testing set IoU and make the predictions more realistic than those of a single model.

However, further investigation is required to understand why the architecture-based ensemble decreases in performance after 8 models, and how a combination of the two ensemble schemes may perform. Additionally, we are also experimenting with regionally-trained models, to further improve smoke detection. In the future, the application of these ensemble techniques are expected to aid in fire and hazard management by automating the monitoring of smoke in real-time from satellite imagery. This tool can be used to provide more frequent and consistent detection of smoke plumes, ultimately supporting climate resilience and adaptation strategies.

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